$$a_n = \frac{1}{n \ln n}$$

Ratio: $\left|\frac{a_{n+1}}{a_n}\right| = \frac{n}{n+1} \cdot \frac{\ln n}{\ln(n+1)}$

$$\frac{n}{n+1} \to 1$$

$$\lim_{n\to\infty} \frac{\ln n}{\ln(n+1)} = \lim_{x\to\infty} \frac{\ln x}{\ln(x+1)} = \lim \frac{\frac{1}{x}}{\frac{1}{x+1}}$$

$$= \lim \frac{x+1}{x} = 1$$

Ex 9.4 Q22

$$\sum \frac{n+1}{n^2\sqrt{n}}$$

$n+1 \sim \underline{n}$

$n^2\sqrt{n} = n^2 n^{1/2}$

$= \underline{n^{5/2}}$

Draft: $a_n = \dfrac{n+1}{n^2\sqrt{n}}$

$b_n = \dfrac{1}{n^{3/2}}$

$\dfrac{n+1}{n^2\sqrt{n}} \sim \dfrac{n}{n^{5/2}} = \underline{\dfrac{1}{n^{3/2}}}$

$\lim \dfrac{a_n}{b_n} = \cdots = 1$

$\sum \dfrac{1}{n^{3/2}}$ conv. $\Leftarrow \dfrac{3}{2} > 1$

[p-series]

Q24

$$\sum \overset{a_n}{\overset{\shortparallel}{\frac{5n^3-3n}{n^2(n-2)(n^2+5)}}} \sim \frac{5n^3}{n^5} = \overset{b_n}{\overset{\shortparallel}{\frac{5}{n^2}}}$$

$2>1$ (Conv.)

Num. $5n^3 - 3n \sim 5n^3$

Denom. $n^2(n-2)(n^2+5) \sim n^2 \cdot n \cdot n^2 = n^5$

$$\lim \frac{a_n}{b_n} = \lim \frac{n^2}{\boxed{5}} \cdot \frac{(5n^3-3n)}{n^2(n-2)(n^2+5)}$$

Q28

$$\sum \frac{(\ln n)^2}{n^3}$$

② $\ln n \le n^{1/2}$

$(\ln n)^2 \le n$

$$\Rightarrow \frac{(\ln n)^2}{n^3} \le \frac{n}{n^3} = \frac{1}{n^2}$$

(Conv.)

① $\ln n \le n$

$\Rightarrow (\ln n)^2 \le n^2$

$$\Rightarrow \frac{(\ln n)^2}{n^3} \le \frac{n^2}{n^3} = \frac{1}{n}$$

Div.

(Don't work)

Q56 $\displaystyle\sum_{n=2}^{\infty} \frac{(\ln n)^2}{n}$

$$= \underbrace{\sum_{n=3}^{\infty} \frac{(\ln n)^2}{n}} + \frac{(\ln 2)^2}{2}$$

$\downarrow$

Div. (DCT)

Const $<<$ $\ln n$

$\downarrow$

$1 \leq \ln n$ for $\underline{n \geq 3}$

$$\frac{(\ln n)^2}{n} \geq \frac{1}{n}$$

$\uparrow$

(Div.)